

Carboxylic Acids; Carboxylic Acid Derivatives; Nucleophilic Acyl Substitution

Organic Chemistry II

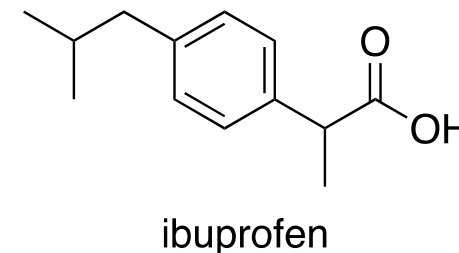
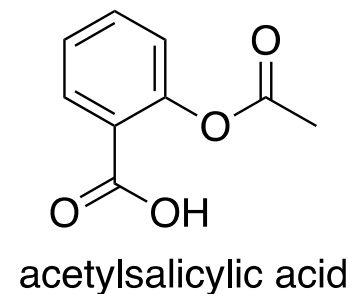
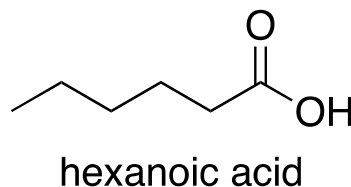
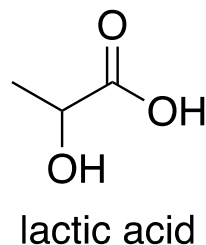
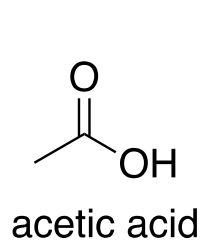
CHEM 2312

Dr. Michael Evans

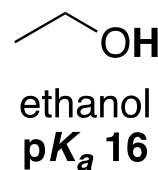
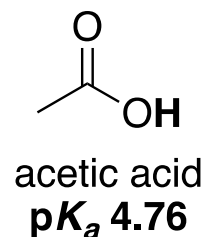
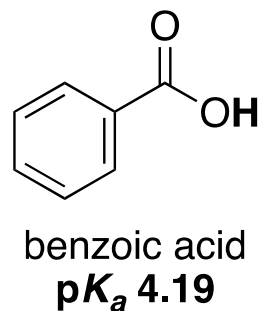
1. Rationalize the acidity of carboxylic acids using resonance and inductive effects.
2. Apply methods for the preparation of carboxylic acids in synthesis.
3. Apply reactions of carboxylic acids in synthesis, including reduction (to aldehydes or alcohols) and oxidation (to acyl chlorides)
4. Describe general mechanisms of nucleophilic acyl substitution (S_NAr) using elementary steps.
5. Predict the products of spontaneous S_NAr reactions and rationalize their spontaneity or lack thereof.
6. Predict the products and describe the mechanisms of “double additions” of organometallic reagents and complex metal hydrides to esters and other carboxylic acid derivatives.
7. Predict the products and describe the mechanisms of hydrolysis reactions of carboxylic acid derivatives.
8. Recognize and apply nitriles (RCN) as carboxylic acid derivatives.

20.1 – 20.3 Structure and Properties of Carboxylic Acids

- **Carboxylic acids** contain the -COOH group (**carboxyl**)
- Abundant both in nature and in synthetic compounds

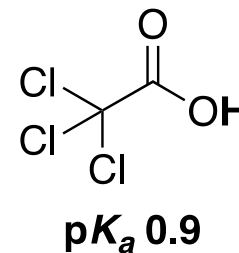
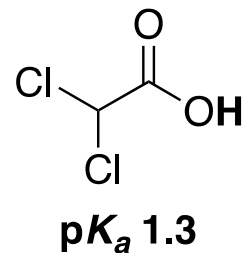
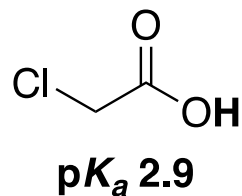
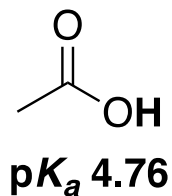


- As organic functional groups go, the carboxyl group is relatively acidic; the conjugate bases of carboxylic acids (**carboxylates**) are stabilized by resonance

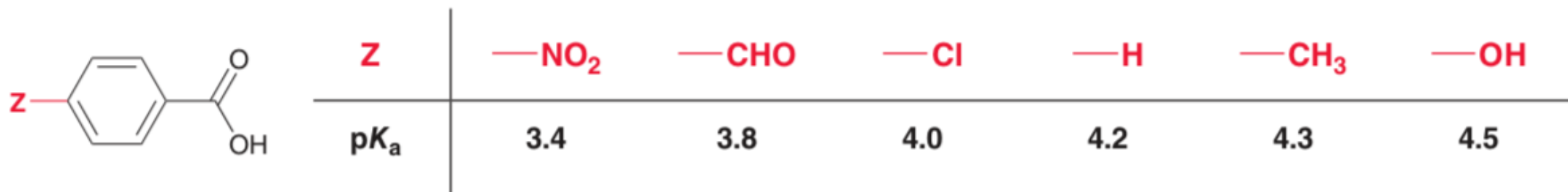


20.3 Structural Effects on Acidity

- Inductively withdrawing substituents (e.g., $-F$, $-Cl$) in the vicinity of a carboxyl group increase its acidity

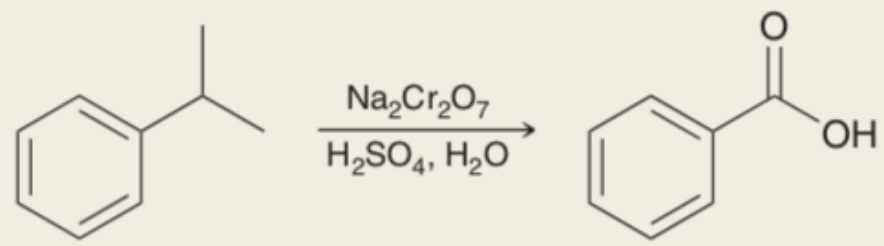


- Electron-withdrawing groups by resonance (e.g., $-NO_2$, $-COR$) “connected” to a carboxyl group increase its acidity

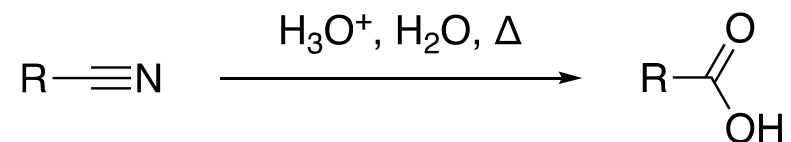


- We have previously seen three methods for the synthesis of carboxylic acids via oxidation

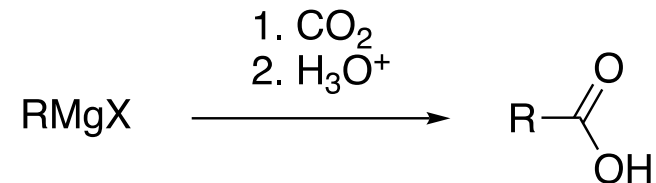
TABLE 20.1 A REVIEW OF METHODS FOR PREPARING CARBOXYLIC ACIDS

REACTION	SECTION NUMBER	COMMENTS
Oxidative Cleavage of Alkynes $\text{R}-\text{C}\equiv\text{C}-\text{R} \xrightarrow[2) \text{H}_2\text{O}]{1) \text{O}_3} \begin{array}{c} \text{HO} \\ \\ \text{R}-\text{C}=\text{O} \end{array} + \begin{array}{c} \text{O} \\ \\ \text{R}-\text{C}-\text{OH} \end{array}$	9.9	Oxidative cleavage will break a $\text{C}\equiv\text{C}$ triple bond forming two carboxylic acids.
Oxidation of Primary Alcohols $\text{R}-\text{CH}_2\text{OH} \xrightarrow[\text{H}_2\text{SO}_4, \text{H}_2\text{O}]{\text{Na}_2\text{Cr}_2\text{O}_7} \text{R}-\text{COOH}$	12.10	A variety of strong oxidizing agents can be used to oxidize primary alcohols and produce carboxylic acids.
Oxidation of Alkylbenzenes 	17.6	Any alkyl group on an aromatic ring will be completely oxidized to give benzoic acid, provided that the benzylic position has at least one hydrogen atom.

- Hydrolysis of nitriles (a nucleophilic acyl substitution reaction...more shortly)

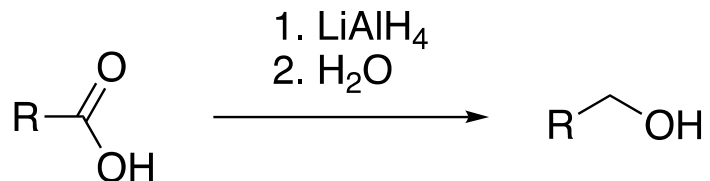


- Addition of a Grignard reagent or organolithium to carbon dioxide

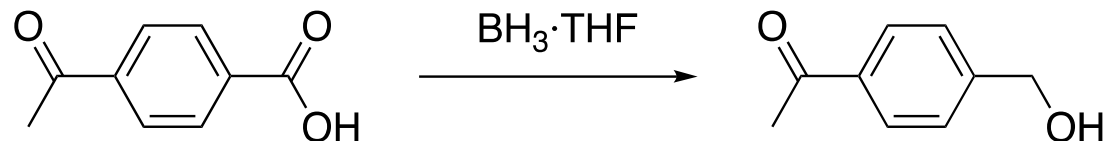


20.5 Reactions of Carboxylic Acids

- Reduction with lithium aluminum hydride (LiAlH_4) to primary alcohols



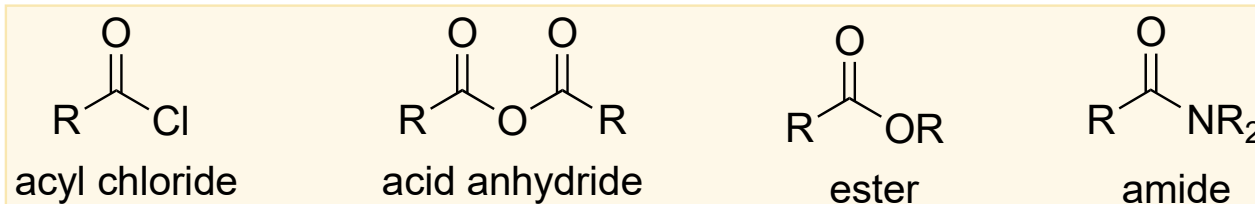
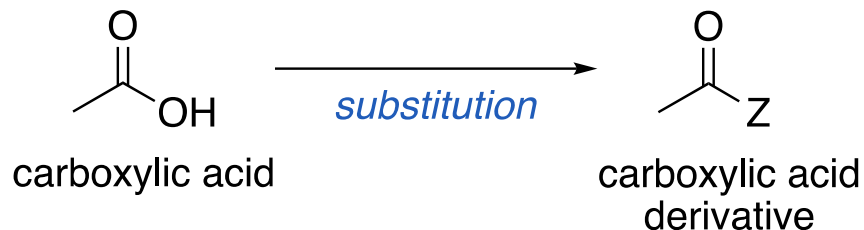
- Selective reduction of carboxylic acids with borane-tetrahydrofuran complex ($\text{BH}_3\cdot\text{THF}$) or $\text{BH}_3\cdot\text{SMe}_2$



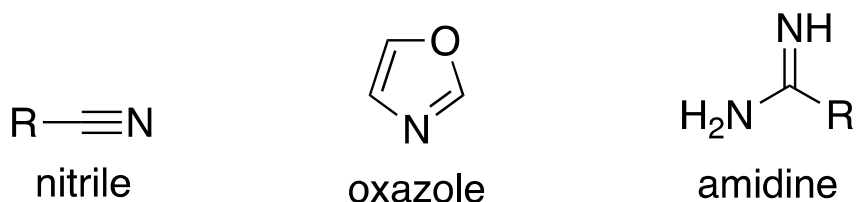
- Conversion to acyl chlorides RCOCl (future slide in this section)
- Peptide coupling (coming soon in CHEM 2312 LS!)

20.6 Introduction to Carboxylic Acid Derivatives

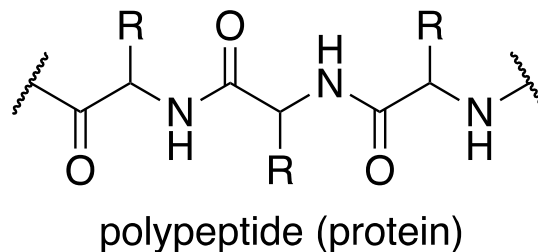
- Carboxylic acid derivatives** are at the same oxidation level as carboxylic acids; the -OH of the carboxyl group is replaced with a different heteroatomic group -Z



- The most important carboxylic acid derivatives contain a carbonyl group...but any group with carbon having three bonds to heteroatoms may be considered a carboxylic acid derivative!

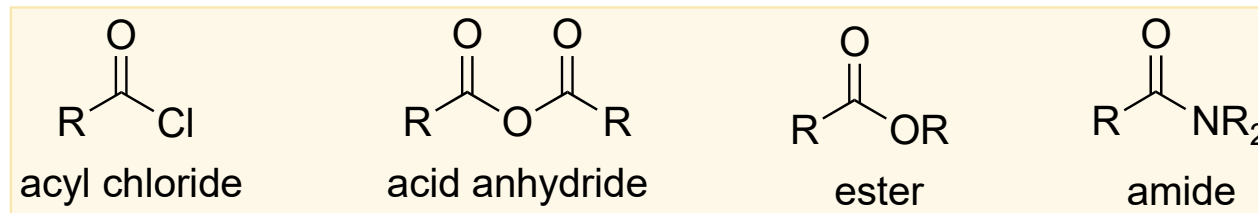


- Carboxylic acid derivatives are ubiquitous in nature as well as synthetic compounds



20.7 Reactivity of Carboxylic Acid Derivatives

- The group $-Z$ influences the electrophilicity of the carbonyl carbon through inductive effects and resonance effects
- Inductive effects.** The $-Z$ atom is withdrawing by induction due to its electronegativity

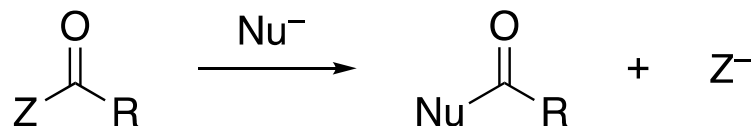


- Resonance effects.** The $-Z$ atom is donating by resonance due to the presence of a lone pair in a p orbital

- Donation by the amide nitrogen is particularly important; amides are planar and rigid!

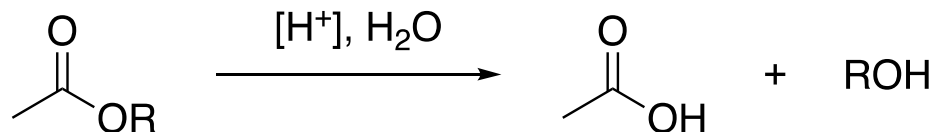
20.7 Nucleophilic Acyl Substitution

- In a **nucleophilic acyl substitution** reaction, a nucleophile replaces Z, which functions as a leaving group



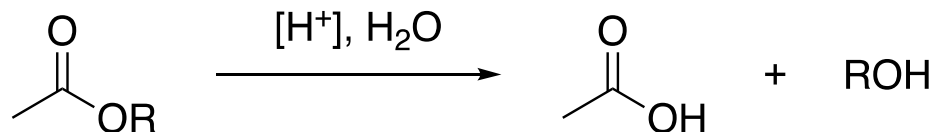
- To be feasible, Z^- must be more stable than Nu^-
- The canonical mechanism involves addition of the nucleophile to the carbonyl carbon (1. Ad_{N}), yielding a **tetrahedral intermediate**, followed by beta-elimination of the leaving group (2. E_{β})

- Under acidic conditions, protonation of the carbonyl group precedes nucleophilic addition (avoids an unreasonable alkoxide intermediate)



- Proton transfer to the carbonyl oxygen occurs first, followed by nucleophilic addition and loss of a proton

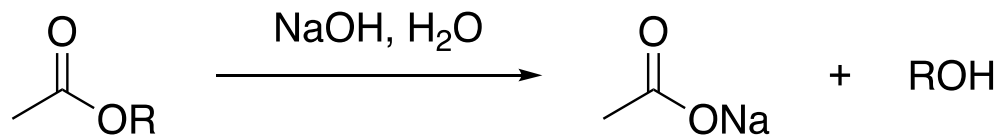
- Under acidic conditions, protonation of the leaving group usually precedes beta-elimination (once again, an unreasonable anionic intermediate is avoided)



- Proton transfer to the alkoxy oxygen occurs first, followed by beta-elimination and loss of a proton

20.7 Nucleophilic Acyl Substitution Under Basic Conditions

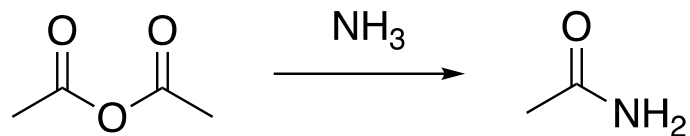
- Under basic conditions, the anionic nucleophile adds directly to the carbonyl group; negative charge is simply transferred from the nucleophile to the carbonyl oxygen



- Nucleophilic addition of hydroxide occurs directly; beta-elimination and proton transfer follow

20.7 Nucleophilic Acyl Substitution by Amine Nucleophiles

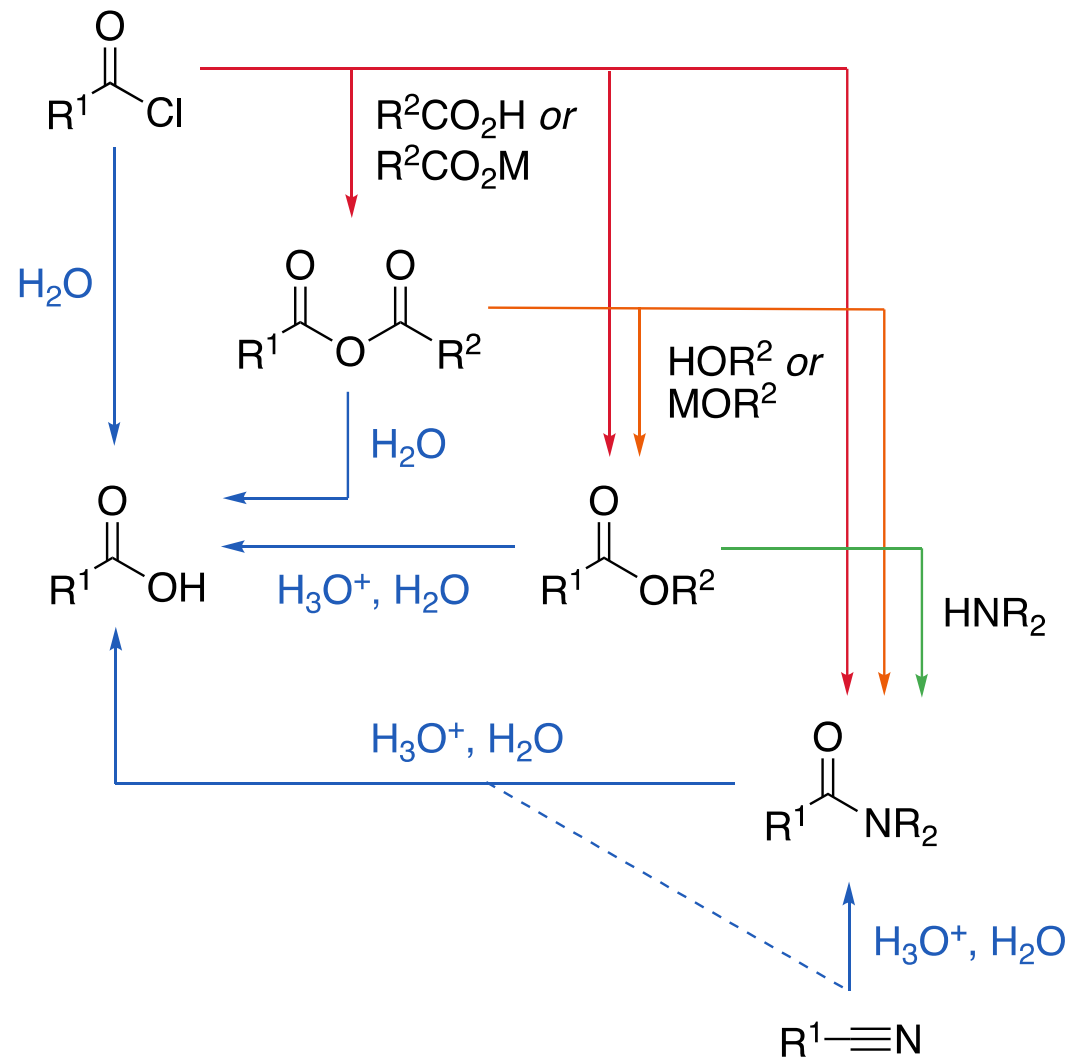
- Invoking zwitterionic intermediates with amine nucleophiles is acceptable



- Ammonia adds directly to the carbonyl carbon without a preceding proton transfer

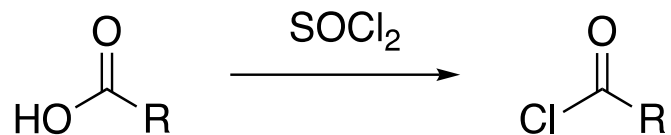
Overview of Nucleophilic Acyl Substitution Reactions

- There are many feasible nucleophilic acyl substitution reactions. You should be able to rationalize and apply all these reactions—don't memorize; understand how and why they work!



20.8 Preparation and Reactions of Acyl Chlorides

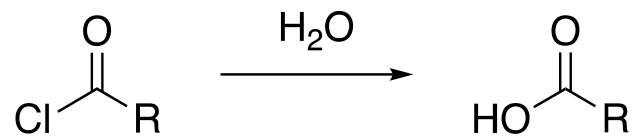
- Acyl chlorides are prepared from carboxylic acids using thionyl chloride (SOCl_2)



- A good leaving group is established at the carbonyl carbon, which is displaced by chloride
- We will see this mechanistic theme again!

20.8 Hydrolysis of Acyl Chlorides (Oopsie!)

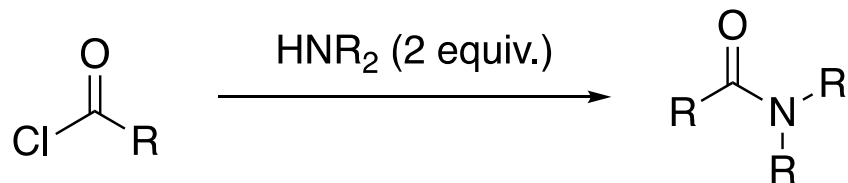
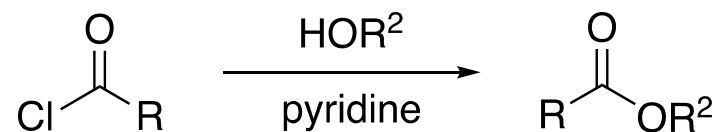
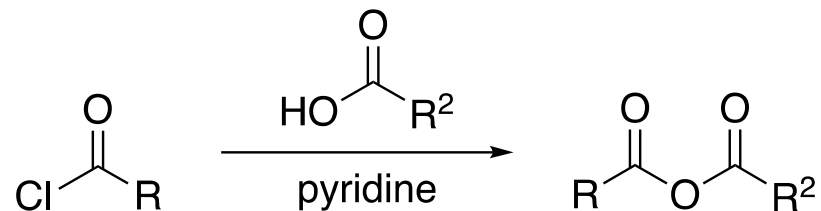
- Acyl chlorides react rapidly with water, even in the open air
- This is not a reaction you'd ever run on purpose



- It *is* a good opportunity to practice drawing the mechanism of nucleophilic acyl substitution: 1. Ad_N; 2. E_β; 3. pt

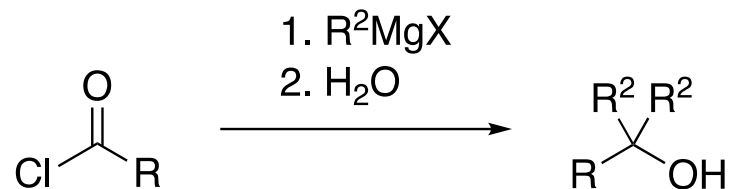
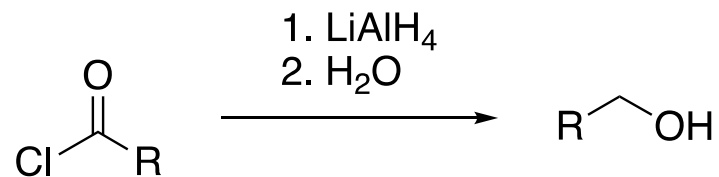
20.8 Substitution Reactions of Acyl Chlorides

- Acyl chlorides react with pretty much any nucleophile to yield a less reactive carboxylic acid derivative
- Pyridine base (or an extra equivalent of HNR_2) is used to sequester HCl generated during the reaction

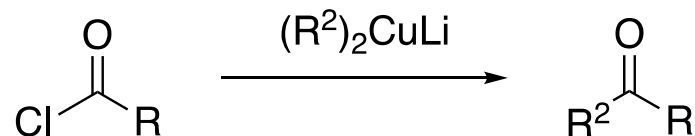


20.8 Reductive Reactions of Acyl Chlorides | 1 of 2

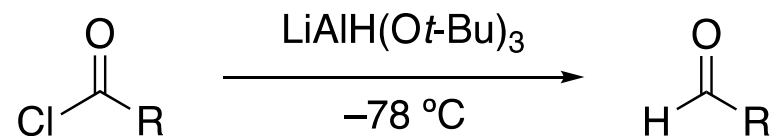
- Acyl chlorides are reduced by lithium aluminum hydride, Grignard reagents, and organolithiums through a “double addition” mechanism



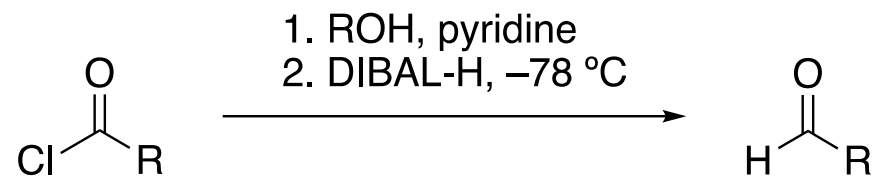
- Lithium organocuprates react with acyl chlorides to yield ketones (we'll learn a superior alternative, though)



- Partial reduction of acyl chlorides to aldehydes can be accomplished using a complex metal hydride with moderated reactivity (inductive withdrawing groups!), $\text{LiAlH}(\text{O}t\text{-Bu})_3$

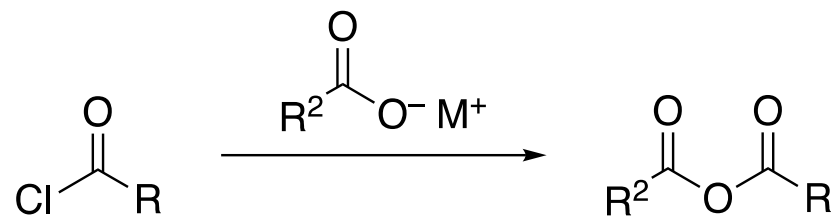


- Conversion to an ester and treatment with DIBAL-H is also an option

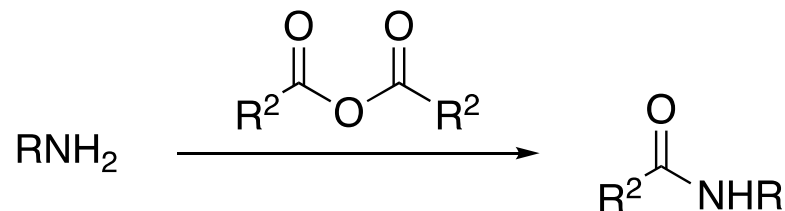
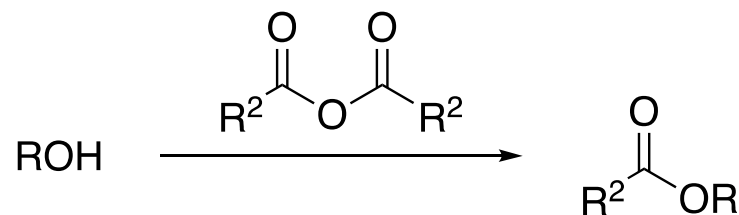
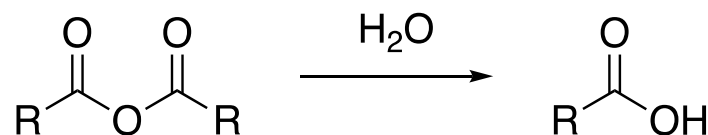


20.9 Preparation and Reactions of Acid Anhydrides

- Acid anhydrides are prepared via treatment of acyl chlorides with carboxylate salts

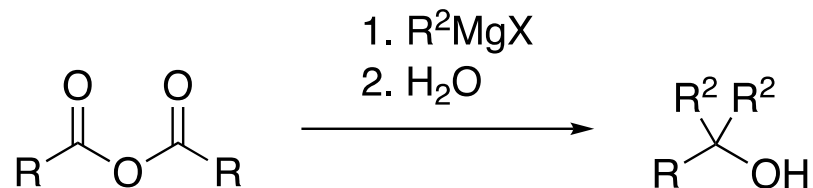
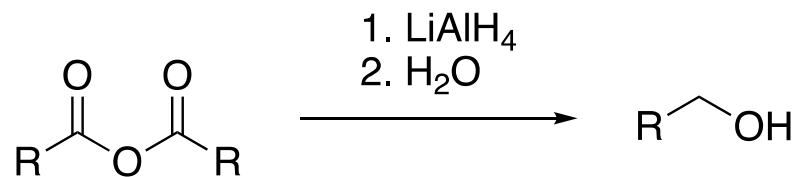


- They undergo substitution in the presence of any nucleophile except chloride; in general, no base is necessary

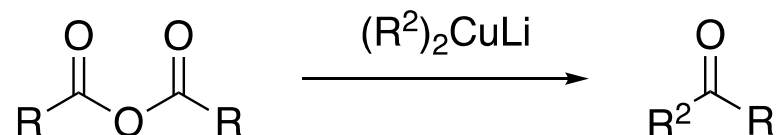


20.9 Reductive Reactions of Acid Anhydrides

- Acid anhydrides are reduced by lithium aluminum hydride, Grignard reagents, and organolithiums through a “double addition” mechanism

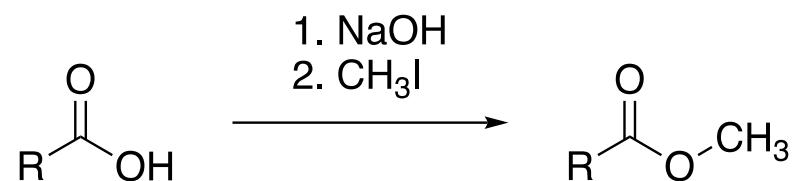


- Lithium organocuprates react with acid anhydrides to yield ketones (we'll learn a superior alternative, though)

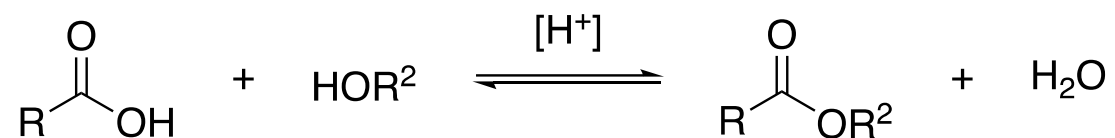


- Partial reduction with $\text{LiAlH}(\text{O}t\text{-Bu})_3$ also works with acid anhydrides

- Esters can be prepared three ways:
 1. Acylation of alcohols with acyl chlorides or anhydrides (see above)
 2. S_N2 reaction of carboxylate nucleophiles

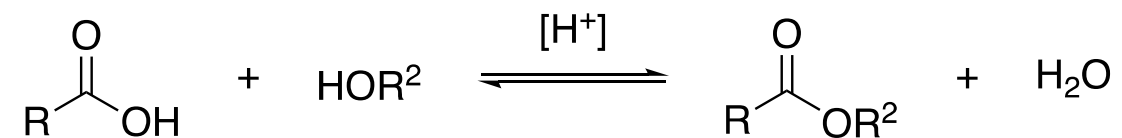


3. Fischer esterification: acid-catalyzed condensation of alcohols and carboxylic acids



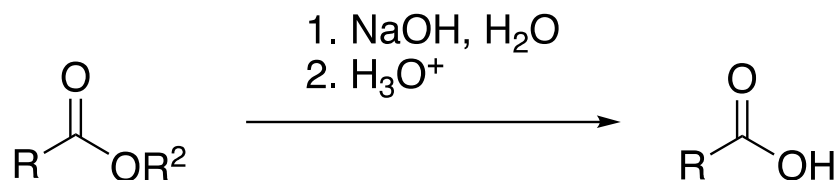
20.10 Mechanism of Fischer Esterification

- The mechanism of Fischer esterification involves acid-catalyzed nucleophilic addition followed by acid-catalyzed beta-elimination



20.11 Reactions of Esters: Saponification

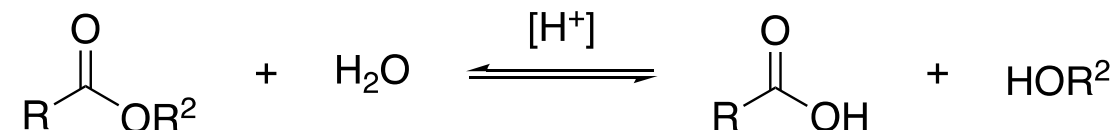
- Because alcohols are generally poor leaving groups, esters have few practically useful substitution reactions
- The base-promoted hydrolysis of esters is called **saponification**



- The mechanism involves nucleophilic acyl substitution followed by an irreversible proton transfer

20.11 Acid-catalyzed Hydrolysis of Esters

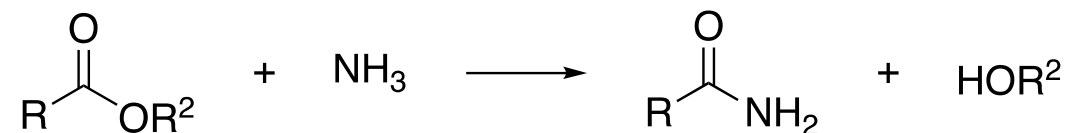
- Treatment of an ester with acidic water results in acid-catalyzed hydrolysis



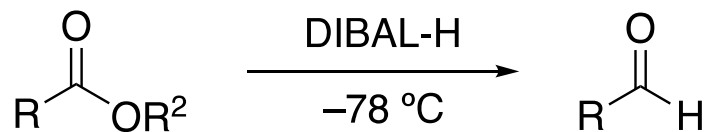
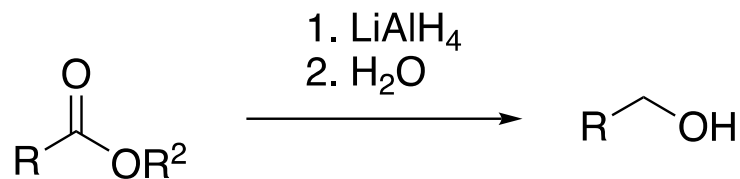
- The mechanism is the microscopic reverse of Fischer esterification: acid-catalyzed nucleophilic addition followed by acid-catalyzed beta-elimination

20.11 Aminolysis and Reductions of Esters

- Although treatment of an ester with an amide does yield an amide slowly, this reaction is rarely used

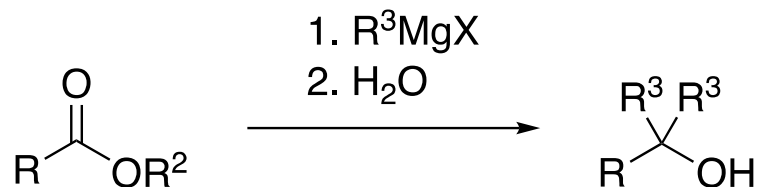


- Esters are reduced to primary alcohols by lithium aluminum hydride (LiAlH_4) and to aldehydes by DIBAL-H



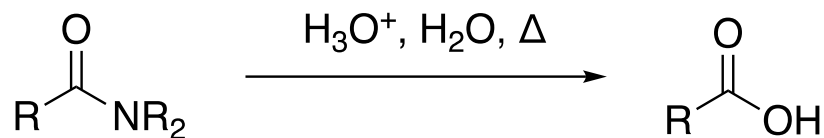
20.11 Double Additions of RMgX and RLi to Esters

- Grignard reagents and organolithiums add twice to esters to yield alcohols via a ketone intermediate

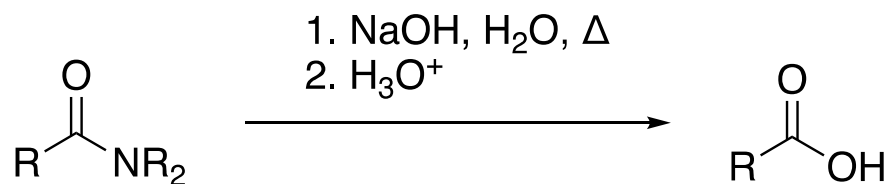


20.12 Preparation and Reactions of Amides

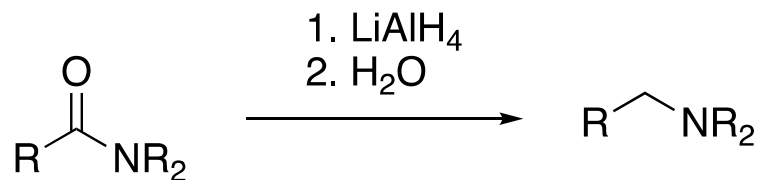
- Use acyl chlorides or anhydrides for preparation of amides (see above)
- Acid-catalyzed hydrolysis of amides is extremely slow but works in a pinch



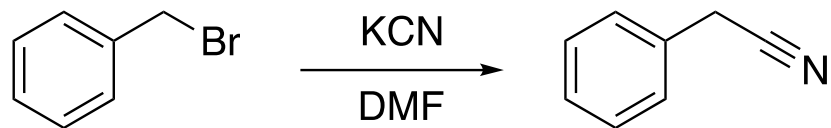
- Hydrolysis under basic conditions is also extremely slow and is less common than the acidic method



- Lithium aluminum hydride (LiAlH_4) reduces amides to amines



1. S_N2 reactions of cyanide anion (CN^-)



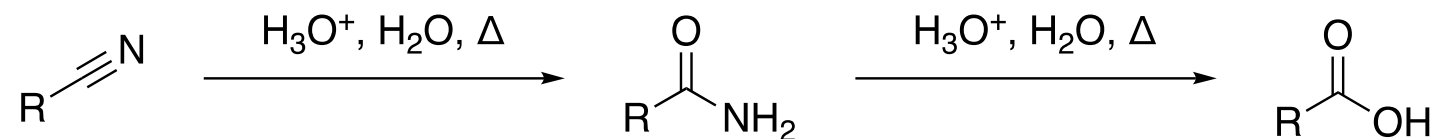
2. Dehydration of primary amides using SOCl_2 or other dehydrating reagents also yields nitriles



- Mechanistically, the carbonyl oxygen is transformed into a good leaving group and eliminated

20.13 Hydrolysis of Nitriles

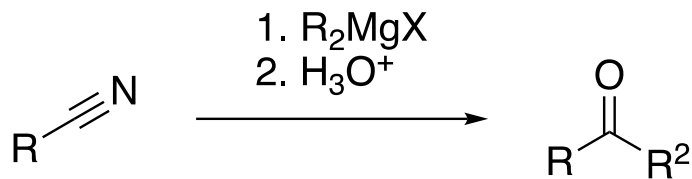
- Nitriles are hydrolyzed in acidic water, first to amides and eventually to carboxylic acids



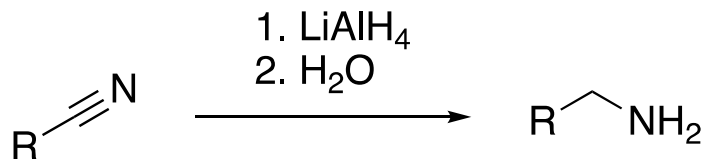
- Acid-catalyzed nucleophilic addition of water followed by tautomerization (two proton transfers)

20.13 Reductive Reactions of Nitriles

- Grignard reagents add to nitriles *only one* to yield an imine; hydrolysis of the imine yields a ketone
- Superior to the organocuprate method for preparation of ketones!



- Nitriles are reduced to primary amines by lithium aluminum hydride (LiAlH_4)



- There are many feasible nucleophilic acyl substitution reactions. You should be able to rationalize and apply all these reactions—don't memorize; understand how and why they work!

